Activity 03 - Train and Evaluate a BigQuery ML Churn Model

Course: Google Professional Machine Learning Engineer Training Course code: C740 Version: v2.0 | 12 September 2026 Evidence: activity artifacts

Objective

Run SQL-defined logistic regression, evaluation and batch prediction on mock customer data.

Deliverables

- customers.csv
- bigquery_ml.sql
- evaluate_metrics.py
- A completed evidence/ folder with decision notes and verification output

Prerequisites

- Python 3.11 or later and a terminal; Docker only where the activity explicitly uses it.
- For optional live Google Cloud steps: the safe training project created in Activity 01.
- Start in this activity folder so relative paths and cleanup commands remain scoped.

Safety, cost and privacy

- Use a training project that contains no production or personal data.
- Replace every placeholder such as <PROJECT_ID> and <REGION> before a cloud command.
- Do not paste API keys or service-account JSON into notebooks, scripts or screenshots.
- Prefer Cloud Shell and Application Default Credentials. Never commit .env or credentials.
- Verify the result against the acceptance checklist, then run the cleanup section.

Evidence to retain

Create an evidence/ folder locally. Save command output, job or revision IDs, metric tables, screenshots without secrets, and a short decision note. Use filenames that include the activity number and date. Evidence is part of the learning outcome, not an optional extra.

Detailed step-by-step guide

Step 1 - Inspect customers.csv; identify the label, entity key and any feature that could leak the outcome

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```
python3 - <<'PY'
import pandas as pd
df=pd.read_csv('customers.csv'); print(df.head()); print(df.isna().sum())
PY
```

Check: Record the observable result or explain the error before continuing.

Step 2 - Open bigquery_ml.sql and replace <PROJECT_ID> and <DATASET> without changing the point-in-time split

Open bigquery_ml.sql and replace <PROJECT_ID> and <DATASET> without changing the point-in-time split.

Check: Record the observable result or explain the error before continuing.

Step 3 - Create the dataset in the same approved region as the training project

Create the dataset in the same approved region as the training project.

```
bq --location=asia-southeast1 mk --dataset "$PROJECT_ID:pmle_lab"
```

Check: Record the observable result or explain the error before continuing.

Step 4 - Load the CSV using the declared schema or autodetect only after inspecting types

Load the CSV using the declared schema or autodetect only after inspecting types.

```
bq load --autodetect --source_format=CSV pmle_lab.customers customers.csv
```

Check: Record the observable result or explain the error before continuing.

Step 5 - Run the CREATE MODEL statement and record the BigQuery job ID and bytes processed

Run the CREATE MODEL statement and record the BigQuery job ID and bytes processed.

```
bq query --use_legacy_sql=false < bigquery_ml.sql
```

Check: Record the observable result or explain the error before continuing.

Step 6 - Query ML.EVALUATE and save accuracy, precision, recall, F1, log loss and ROC AUC

Query ML.EVALUATE and save accuracy, precision, recall, F1, log loss and ROC AUC.

Check: Record the observable result or explain the error before continuing.

Step 7 - Run evaluate_metrics.py to compare the supplied confusion matrices at three thresholds

Run evaluate_metrics.py to compare the supplied confusion matrices at three thresholds.

```
python3 evaluate_metrics.py
```

Check: Record the observable result or explain the error before continuing.

Step 8 - Choose a threshold using the stated false-negative cost, not accuracy alone, and document the rationale

Choose a threshold using the stated false-negative cost, not accuracy alone, and document the rationale.

Check: Record the observable result or explain the error before continuing.

Step 9 - Run ML.PREDICT for the holdout rows and check that no post-outcome feature enters the model

Run ML.PREDICT for the holdout rows and check that no post-outcome feature enters the model.

Check: Record the observable result or explain the error before continuing.

Step 10 - Delete the model and dataset during cleanup if the trainer does not require them for a later activity

Delete the model and dataset during cleanup if the trainer does not require them for a later activity.

```
bq rm -f -m pmle_lab.churn_model
```

Check: Record the observable result or explain the error before continuing.

Acceptance checklist

[ ] Every required artifact exists and opens without error.

[ ] Results are tied to the supplied mock data or an approved training project.

[ ] The conclusion cites metrics, IDs or command output rather than opinion alone.

[ ] No credentials, tokens, personal data or unredacted sensitive prompts are present.

[ ] Cleanup is completed or the retained resource is documented with an owner and expiry.

Troubleshooting

Symptom	Likely cause	Corrective action
403 PERMISSION_DENIED	API disabled, wrong project, or IAM role missing	Verify gcloud config get-value project, API state and the least-privilege role.
429 RESOURCE_EXHAUSTED	Quota or rate limit reached	Stop retries, inspect quota, add exponential backoff, and ask the trainer before increasing quota.
Region/location mismatch	Dataset, bucket, model or endpoint uses another location	Recreate only the training resource in one approved region; do not copy sensitive data casually.
Result differs from expected	Data, dependency or configuration version drifted	Compare file hashes, pinned versions and parameters; retain the diff as evidence.

Cleanup

Stop local processes and remove only resources created for this activity. For live cloud work, use the Console resource views and activity evidence to verify deletion. Do not delete a shared project, dataset, bucket, registry or endpoint unless the trainer identifies it as disposable.